

DATABASE MANAGEMENT SYSTEMS (CS301)

SQL Basics

Learning Objectives

Understand SQL fundamentals, data types, DDL, DML, DCL, TCL commands, and query writing techniques.

Introduction to SQL

Structured Query Language (SQL) is the standard language used to create, manage, and manipulate relational databases.

Features of SQL

- Easy to learn and use
- Supports data definition and manipulation
- Provides security controls
- Supports transaction management

Common SQL Data Types

- INT – Integer values
- VARCHAR – Variable length strings
- CHAR – Fixed length strings
- DATE – Date values
- FLOAT – Decimal numbers

DDL Commands

Data Definition Language (DDL) commands define and modify database structures.

CREATE Command

Used to create tables and databases. Example: `CREATE TABLE Student(StudentID INT, Name VARCHAR(50));`

ALTER Command

Used to modify existing table structures. Example: `ALTER TABLE Student ADD Branch VARCHAR(20);`

DROP Command

Used to remove tables or databases. Example: `DROP TABLE Student;`

DML Commands

Data Manipulation Language (DML) commands are used to insert, update, and delete records.

INSERT Command

`INSERT INTO Student VALUES(101, 'Rahul');`

UPDATE Command

UPDATE Student SET Name='Ravi' WHERE StudentID=101;

DELETE Command

DELETE FROM Student WHERE StudentID=101;

SELECT Query

SELECT * FROM Student;

WHERE Clause

Used to filter records. Example: SELECT * FROM Student WHERE Branch='CSE';

ORDER BY Clause

Used to sort records. Example: SELECT * FROM Student ORDER BY Name;

GROUP BY Clause

Used with aggregate functions. Example: SELECT Branch, COUNT(*) FROM Student GROUP BY Branch;

DCL Commands

GRANT – Provides permissions.

REVOKE – Removes permissions.

TCL Commands

COMMIT – Saves changes.

ROLLBACK – Undoes changes.

Applications of SQL

- Banking Systems
- E-Commerce Platforms
- Hospital Management Systems
- University Databases

Exam Important Questions

1. Define SQL.
2. Explain DDL commands.
3. Explain DML commands.
4. Differentiate DDL and DML.
5. Write SQL queries using SELECT and WHERE.

Summary

SQL is the standard language for relational databases. Understanding SQL commands and query writing is essential for database development and administration.